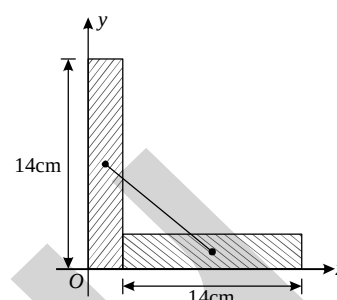


Section - A : MCQs with Single Option Correct

1. A metal sheet $14\text{ cm} \times 2\text{ cm}$ of uniform thickness is cut into two pieces of width 1 cm each. The two pieces are joined and laid along xy plane as shown. The centre of mass has the coordinates :

- (A) $(1, 1)$
 (B) $\left(\frac{19}{7}, \frac{19}{7}\right)$
 (C) $\left(\frac{8}{7}, \frac{8}{7}\right)$

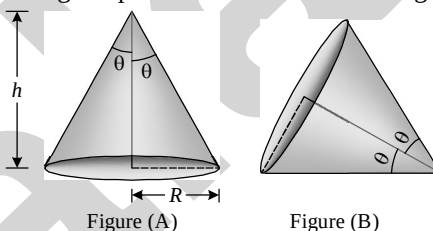
(D) None of these



2. Three point masses, $m_1 = 2\text{ kg}$, $m_2 = 4\text{ kg}$ and $m_3 = 6\text{ kg}$ are kept at the three corners of an equilateral triangle of side 1 m . The location of their center of mass is :

- (A) $\left(\frac{7}{12}, \frac{\sqrt{3}}{4}\right)$ (B) $\left(\frac{5}{12}, \frac{\sqrt{5}}{4}\right)$ (C) $\left(\frac{9}{7}, \frac{\sqrt{7}}{6}\right)$ (D) $\left(\frac{11}{12}, \frac{\sqrt{3}}{8}\right)$

3. A solid cone is placed on a horizontal surface has height h , radius R and apex angle θ as shown. If gravitational potential energy of the cone does not change as position of the cone is changed from figure (A) to figure (B), then,



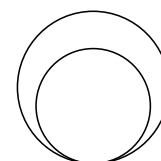
- (A) $\theta = \sin^{-1}\left(\frac{2}{3}\right)$ (B) $\theta = \frac{\pi}{4}$ (C) $\frac{h}{R} = 1$ (D) $\frac{h}{R} = 2\sqrt{2}$

4. Two particles P and Q are in motion under gravity. Then, identify the incorrect statement :

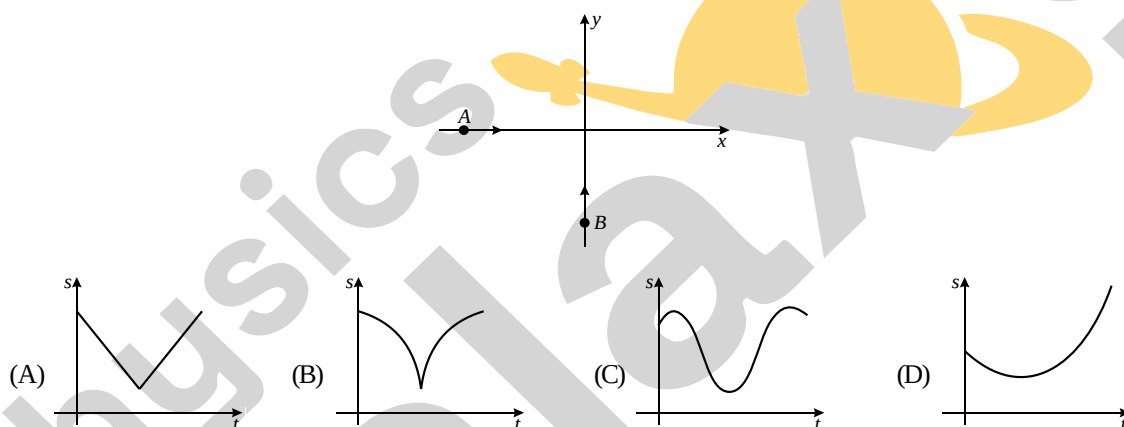
- (A) Their relative velocity may not be zero
 (B) Those falls through equal heights in equal segment of time
 (C) Their CM does not have constant velocity
 (D) Their CM has constant acceleration

5. From a thin circular disc of radius R , a circular hole of radius $\frac{4R}{5}$ is cut as shown in figure. The distance of centre of mass of remaining disc from the centre of original disc is :

- (A) $\frac{15R}{40}$ (B) $\frac{R}{3}$
 (C) $\frac{R}{4}$ (D) None of these



6. Two spherical bodies of mass M and $5M$ and radii R and $2R$ respectively are released in free space with initial separation between their centres equal to $12R$. If they attract each other due to gravitational force only, then the distance covered by the smaller body just before collision is :
 (A) $2.5R$ (B) $4.5R$ (C) $7.5R$ (D) $1.5R$
7. A body A of mass M while falling vertically downwards under gravity breaks into two parts; a body B of mass $M/3$ and, a body C of mass $2M/3$. The centre of mass of bodies B and C taken together shifts compared to that of body A towards :
 (A) Depends on height of breaking (B) Does not shift
 (C) Body C (D) Body B
8. A mass m moves with a velocity v and collides inelastically with another identical mass. After collision the 1st mass moves with velocity $v/\sqrt{3}$ in a direction perpendicular to the initial direction of motion. Find the speed of the second mass after collision :
 (A) v (B) $\sqrt{3}v$ (C) $\frac{2}{\sqrt{3}}v$ (D) $\frac{v}{\sqrt{3}}$
9. Particle A and B are moving with constant velocities along positive x and positive y -axis respectively as shown in figure. The graph of separation between them with time is represented by (assume that they will not collide)



10. A small mass m is launched from the top of a cliff with speed V at an angle of 30° above the horizontal. When the mass reaches the ground, its velocity is directed at 45° below the horizontal. Which one of the following choices is the magnitude of the total impulse that was imparted to the mass during its flight? Ignore air resistance.
 (A) $\frac{1}{2}(\sqrt{3}+1)mV$ (B) $\frac{\sqrt{3}}{2}\left(\frac{\sqrt{2}+1}{2}\right)mV$ (C) $\frac{1}{2}(\sqrt{3}-1)mV$ (D) $\frac{1}{2}\left(\frac{\sqrt{3}}{2}+1\right)mV$
11. In figure, determine the character of the collision. The masses of the blocks, and the velocities before and after are given. The collision is :
 (A) Perfectly elastic
 (B) Partially inelastic
 (C) Completely inelastic
 (D) This collision is not possible
- Before: $\begin{array}{|c|c|} \hline 4 \text{ kg} & 6 \text{ kg} \\ \hline \end{array}$ $\begin{array}{c} 1.8 \text{ m/s} \rightarrow \\ \leftarrow 0.2 \text{ m/s} \end{array}$

After: $\begin{array}{|c|c|} \hline 4 \text{ kg} & 6 \text{ kg} \\ \hline \end{array}$ $\begin{array}{c} \leftarrow 0.6 \text{ m/s} \\ 1.4 \text{ m/s} \rightarrow \end{array}$
12. Consider a one-dimensional collision where a body of mass m_1 originally moving in the positive x -direction with speed v_0 collides with a second body of mass m_2 originally at rest. The collision could be completely inelastic, with the two bodies sticking together, completely elastic, or somewhere in between. After the collision, m_1 moves with velocity $\vec{v}_1$, while m_2 moves with velocity $\vec{v}_2$. If $m_1 > m_2$, then :

- (A) $v_0 < v_1 < v_2$ (B) $v_2 < v_1 < v_0$ (C) $\frac{v_0}{2} < v_2 < v_1$ (D) $\frac{v_0}{2} < v_2 < 2v_0$

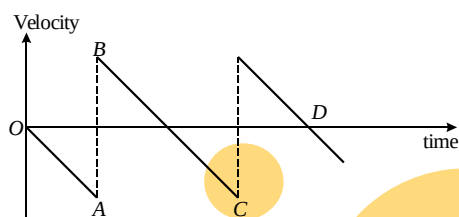
13. A ball of mass m hits directly another ball of mass M at rest and is brought to rest by the impact. One third of the kinetic energy of the ball is lost due to the collision. The coefficient of restitution is :

(A) $1/3$ (B) $1/2$ (C) $2/3$ (D) $\sqrt{\frac{2}{3}}$

14. A ball A of mass 1 kg moving at a speed of 5 m/s strikes tangentially another ball B initially at rest. The ball A then moves at right angles to its initial direction at a speed of 4 m/s. If the collision is elastic, the mass (in kg) of ball B and its momentum after collision (in kg-m/s) respectively are approximately

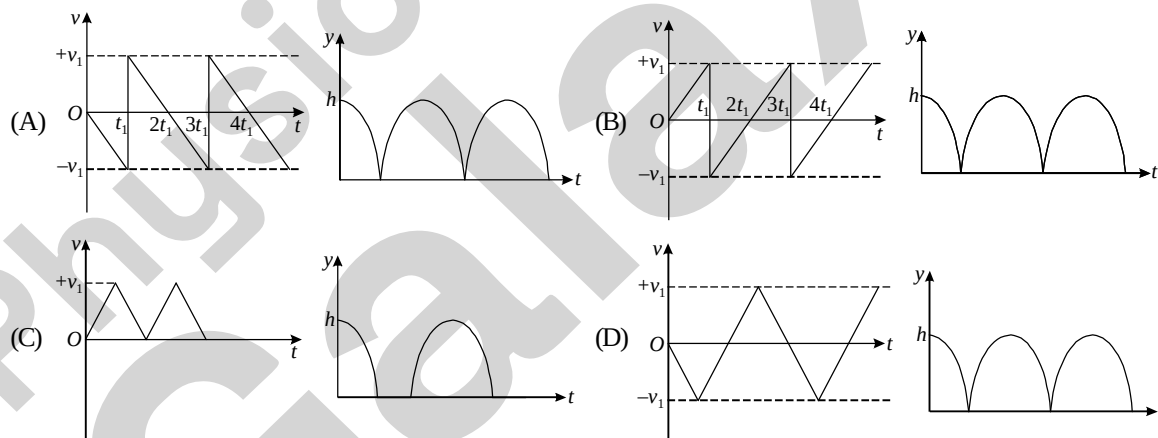
(A) 1.2 and 1.8 (B) 2.2 and 3.3 (C) 4.6 and 6.4 (D) 6.2 and 9.1

15. A tennis ball is released from a height and allowed to fall onto a hard surface. The adjacent graph shows the variation of velocity of the ball with time from the instant of release. The point of upward maximum velocity of the ball is indicated by point :

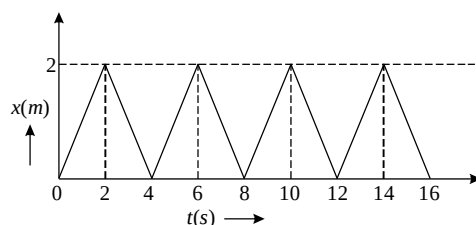


(A) A (B) B (C) C (D) D

16. Consider a rubber ball freely falling from a height $h = 4.9$ m onto a horizontal elastic plate. Assume that the duration of collision is negligible and the collision with the plate is totally elastic. Then the velocity as a function of time and the height as a function of time will be :



17. The figure shows the position-time ($x-t$) graph of one dimensional motion of a body of mass 0.4 kg. The magnitude of each impulse at change of velocity is given as

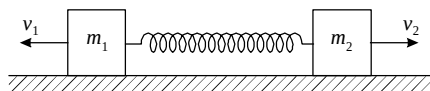


(A) 0.4 Ns (B) 0.8 Ns (C) 1.6 Ns (D) 0.2 Ns

18. A particle of mass 'm' is projected with a velocity v making an angle of 30° with the horizontal. The magnitude of angular momentum of the projectile about the point of projection when the particle is at its maximum height 'h' is :

(A) $\frac{\sqrt{3}}{2} \frac{mv^2}{g}$ (B) Zero (C) $\frac{mv^3}{\sqrt{2}g}$ (D) $\frac{\sqrt{3}}{16} \frac{mv^3}{g}$

19. A spring is compressed between two blocks of masses m_1 and m_2 placed on a horizontal frictionless surface as shown in the figure. When the blocks are released, they have initial velocity of v_1 and v_2 as shown. The block travel distances x_1 and x_2 respectively before coming to rest. The ratio (x_1/x_2) is :



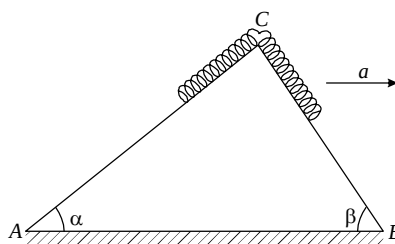
(A) $\frac{m_2}{m_1}$ (B) $\frac{m_1}{m_2}$ (C) $\sqrt{\frac{m_2}{m_1}}$ (D) $\sqrt{\frac{m_1}{m_2}}$

20. Two points of a rod move with velocities $3v$ & v perpendicular to the rod and in the same direction, separated by a distance 'r'. Then the angular velocity of the rod is :

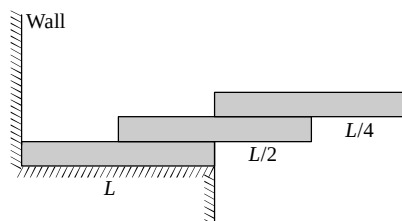
(A) $\frac{3v}{r}$ (B) $\frac{4v}{r}$ (C) $\frac{5v}{r}$ (D) $\frac{2v}{r}$

INTEGRITYTYPE

21. A block of mass 2 kg is connected to another block of mass 1 kg by a spring (massless) of spring constant k . The blocks are kept on a smooth horizontal plane. Initially the blocks are at rest and the spring is unstretched. Then a constant force of 15 N starts acting on the block of mass 1 kg to pull it. Find the net force (N) on the block of mass 2 kg in centre of mass frame, initially.
22. A shell flying with velocity $v = 500$ m/s bursts into three identical fragments so that the kinetic energy of the system increases $\eta = 1.5$ times. What maximum velocity can one of the fragments obtain in km/s?
23. A bomb of mass 16 kg at rest explodes into two pieces of masses 4 kg and 12 kg. The velocity of the 12 kg mass is 4 ms^{-1} . Calculate the kinetic energy (in Joule) of the other mass.
24. A block of mass 0.50 kg is moving with a speed of 2.00 ms^{-1} on a smooth surface. It strikes another mass of 1.00 kg and then they move together as a single body. The energy (in joule) loss during the collision is $20/k$ find k .
25. The time dependence of the position of a particle of mass $m = 2$ kg is given by $\vec{r}(t) = 2t\hat{i} - 3t^2\hat{j}$ m. Calculate its angular momentum in $\text{kg-m}^2/\text{s}$, with respect to the origin, at time $t = 2$ s.
26. Two identical particles move towards each other with velocity $2v$ and v respectively. The velocity of centre of mass is v/m then calculate m .
27. A uniform chain is placed on a wedge ABC with its midpoint lying on vertex C. The acceleration to be imparted to the wedge to the right so that the chain remains stationary with respect to the wedge is $a = \frac{g}{n}$, where $n = \underline{\hspace{1cm}}$ (Neglect friction; take $\cos \alpha = \frac{4}{5}$, $\cos \beta = \frac{3}{5}$).



28. Three bricks each of length L and mass M are arranged, as shown from the wall. The distance of the centre of mass of the system from the wall is $\frac{nL}{24}$ find n .



29. Density of a sphere varies as $\rho = \rho_0 + 2kr^2$. r is the distance from the centre of the sphere. Find the distance of its centre of mass from the centre of the sphere.
30. A ball falls vertically for 2 seconds and hits a smooth plane inclined at 30° to the horizontal. If the coefficient of restitution is $3/4$, find the time in seconds that elapses before it again hits the plane.

* * * * *

Topic : SYSTEM OF PARTICLES**ANSWER KEY****Section - A : MCQs with Single Option Correct**

- | | | | |
|---------|---------|---------|---------|
| 1. (D) | 2. (A) | 3. (D) | 4. (C) |
| 5. (D) | 6. (C) | 7. (B) | 8. (C) |
| 9. (D) | 10. (A) | 11. (A) | 12. (D) |
| 13. (C) | 14. (C) | 15. (B) | 16. (A) |
| 17. (B) | 18. (D) | 19. (A) | 20. (D) |

Section- B: INTEGER Answer Type Questions

- | | | | |
|----------|---------|-------------|----------|
| 21. [10] | 22. [1] | 23. [288 J] | 24. [30] |
| 25. [48] | 26. [2] | 27. [7] | 28. [22] |
| 29. [0] | 30. [3] | | |